2024 CpSc Q5

Section: Computer Systems

Topic: Cache Memory — Improving Performance

Question Summary

Computer performance is improved by the inclusion of cache memory on the same computer chip as the processor. Describe how cache memory improves performance.

Worked Solution

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Cache memory is a small but very fast type of memory built close to, or directly inside, the CPU. It stores copies of instructions and data that the CPU is likely to reuse soon.

When the processor needs data:
• It first checks whether the data is already in the cache.
• If found (a cache hit), it can be accessed much faster than if it had to be fetched from the main memory (RAM).
• If not found (a cache miss), the data must be read from RAM and placed in the cache for future use.

Because cache memory operates at or near CPU speed, it greatly reduces the average time needed for the processor to access frequently used instructions and data, thus improving performance.

Final Answer

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Cache memory improves performance by storing frequently used instructions and data close to the processor, reducing the time the CPU spends waiting for data from main memory.

Revision Tips

- Cache is faster but smaller than RAM.
- The CPU checks cache before RAM — a cache hit speeds up access.
- Modern CPUs often have several cache levels (L1, L2, L3).
- Performance improvement comes from reduced memory access time, not from increased clock speed.

Exam Alignment

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This explanation matches the 2024 marking instructions: candidates should describe that cache stores frequently used data/instructions close to the processor, reducing access time and therefore improving performance.